

## GARTLAND, Norway

WMO: 012910

Lat: 64.531N

Lon: 12.384E

Elev: 95

StdP: 100.19

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -22.8      | -19.2 | -25.1                         | 0.4 | -22.5 | -21.5 | 0.6 | -18.8 | 10.5                  | -4.2 | 9.4 | -3.8 | 0.9                   | 240  | 0.644 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.7                   | 26.3            | 17.6 | 23.8 | 16.3 | 21.7 | 15.2 | 18.6                | 24.4 | 17.2 | 22.3 | 16.0 | 20.6 | 2.6                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.2                            | 11.6 | 20.7 | 15.0 | 10.8 | 19.2 | 13.9 | 10.0 | 17.5 | 52.9          | 24.3 | 48.5 | 22.3 | 45.1 | 20.8 | 23.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 7.6               | 6.4  | 5.4 | -24.1                      | 28.7 | 4.5                | 2.8 | -27.4                                              | 30.7 | -30.1      | 32.4 | -32.6      | 34.0 | -35.9      | 36.0 |
|                   |      |     | -24.2                      | 20.1 | 4.4                | 1.8 | -27.4                                              | 21.4 | -29.9      | 22.4 | -32.4      | 23.4 | -35.6      | 24.7 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 3.8    | -5.2  | -4.8  | -1.8  | 2.7   | 7.5   | 11.2  | 14.5  | 12.9  | 9.4   | 3.7   | -1.3  | -4.0  |
|                                                                                  | DBStd       | 8.22   | 6.16  | 5.88  | 4.83  | 3.02  | 4.03  | 3.56  | 3.54  | 2.67  | 2.96  | 3.45  | 5.39  | 6.09  |
|                                                                                  | HDD10.0     | 2619   | 472   | 414   | 366   | 219   | 101   | 25    | 3     | 4     | 44    | 197   | 339   | 434   |
|                                                                                  | HDD18.3     | 5326   | 730   | 648   | 624   | 469   | 336   | 217   | 128   | 171   | 267   | 453   | 589   | 693   |
|                                                                                  | CDD10.0     | 349    | 0     | 0     | 0     | 0     | 24    | 61    | 142   | 93    | 27    | 2     | 0     | 0     |
|                                                                                  | CDD18.3     | 13     | 0     | 0     | 0     | 0     | 0     | 2     | 9     | 1     | 0     | 0     | 0     | 0     |
|                                                                                  | CDH23.3     | 259    | 0     | 0     | 0     | 0     | 17    | 46    | 169   | 28    | 0     | 0     | 0     | 0     |
|                                                                                  | CDH26.7     | 57     | 0     | 0     | 0     | 0     | 1     | 8     | 47    | 1     | 0     | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 2.1    | 2.4   | 2.4   | 2.3   | 2.0   | 2.2   | 2.0   | 1.9   | 1.7   | 1.9   | 2.0   | 1.9   | 2.1   |
| Precipitation                                                                    | PrecAvg     | 1304   | 135   | 115   | 116   | 78    | 64    | 83    | 92    | 93    | 128   | 131   | 121   | 149   |
|                                                                                  | PrecMax     | 1712   | 300   | 293   | 265   | 168   | 92    | 165   | 202   | 159   | 314   | 214   | 254   | 273   |
|                                                                                  | PrecMin     | 893    | 14    | 13    | 18    | 28    | 26    | 7     | 42    | 37    | 33    | 43    | 12    | 45    |
|                                                                                  | PrecStd     | 193    | 73    | 66    | 66    | 39    | 19    | 38    | 37    | 34    | 66    | 46    | 70    | 69    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 6.0   | 6.7   | 8.9   | 18.0  | 25.3  | 27.9  | 31.1  | 26.1  | 21.5  | 15.9  | 9.7   | 7.4   |
|                                                                                  |             | MCWB   | 5.6   | 4.7   | 4.6   | 10.2  | 15.6  | 17.0  | 20.1  | 17.6  | 15.2  | 11.6  | 9.0   | 5.5   |
|                                                                                  | 2%          | DB     | 4.4   | 5.3   | 6.4   | 14.2  | 22.4  | 23.9  | 27.6  | 23.8  | 18.9  | 12.5  | 7.9   | 5.7   |
|                                                                                  |             | MCWB   | 3.5   | 4.0   | 4.1   | 7.8   | 13.8  | 15.4  | 18.8  | 16.6  | 14.1  | 9.6   | 6.8   | 4.5   |
|                                                                                  | 5%          | DB     | 3.2   | 4.3   | 5.3   | 10.7  | 19.5  | 21.2  | 24.8  | 21.6  | 16.8  | 10.6  | 6.0   | 4.2   |
|                                                                                  |             | MCWB   | 2.2   | 3.1   | 3.6   | 5.9   | 12.4  | 14.3  | 17.4  | 15.8  | 13.0  | 8.8   | 4.5   | 3.0   |
|                                                                                  | 10%         | DB     | 1.6   | 2.8   | 4.2   | 8.2   | 15.9  | 18.7  | 22.4  | 19.4  | 15.0  | 8.7   | 4.7   | 3.1   |
|                                                                                  |             | MCWB   | 1.0   | 1.6   | 2.6   | 4.6   | 10.4  | 13.2  | 16.3  | 15.0  | 12.0  | 7.4   | 3.6   | 1.9   |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 5.5   | 5.3   | 5.8   | 10.5  | 16.3  | 17.7  | 21.8  | 18.9  | 16.0  | 11.8  | 9.3   | 6.6   |
|                                                                                  |             | MCDB   | 5.8   | 6.0   | 6.6   | 17.4  | 24.5  | 25.3  | 29.7  | 23.0  | 19.4  | 14.7  | 9.8   | 6.9   |
|                                                                                  | 2%          | WB     | 3.6   | 4.3   | 4.7   | 8.2   | 14.5  | 16.5  | 19.7  | 17.5  | 14.7  | 10.1  | 7.1   | 4.6   |
|                                                                                  |             | MCDB   | 4.1   | 5.2   | 5.9   | 13.7  | 21.0  | 22.9  | 26.1  | 21.7  | 18.0  | 11.9  | 7.8   | 5.2   |
|                                                                                  | 5%          | WB     | 2.2   | 3.2   | 3.7   | 6.5   | 12.6  | 15.1  | 18.2  | 16.3  | 13.5  | 8.9   | 4.8   | 3.1   |
|                                                                                  |             | MCDB   | 2.9   | 4.1   | 5.1   | 9.7   | 18.6  | 19.8  | 23.9  | 20.6  | 16.2  | 10.3  | 5.6   | 4.0   |
|                                                                                  | 10%         | WB     | 1.0   | 1.5   | 2.7   | 5.1   | 10.9  | 13.7  | 16.7  | 15.3  | 12.3  | 7.6   | 3.7   | 2.0   |
|                                                                                  |             | MCDB   | 1.5   | 2.5   | 3.9   | 7.4   | 15.3  | 17.7  | 21.3  | 18.9  | 14.6  | 8.5   | 4.5   | 2.8   |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.8   | 6.9   | 8.3   | 9.8   | 11.0  | 10.1  | 10.7  | 9.3   | 6.7   | 5.2   | 6.0   |       |
|                                                                                  |             | MCDBR  | 5.0   | 6.1   | 7.8   | 15.4  | 17.4  | 15.6  | 16.3  | 15.4  | 12.8  | 8.9   | 5.9   | 6.2   |
|                                                                                  |             | MCWBR  | 4.3   | 4.9   | 5.7   | 9.4   | 9.5   | 8.0   | 8.3   | 8.9   | 6.4   | 5.1   | 5.3   |       |
|                                                                                  | 5% WB       | MCDBR  | 4.8   | 5.7   | 6.8   | 13.2  | 15.8  | 13.3  | 14.8  | 13.6  | 11.9  | 7.8   | 5.7   | 5.9   |
|                                                                                  |             | MCWBR  | 4.4   | 5.0   | 5.4   | 8.5   | 8.9   | 7.0   | 7.8   | 8.4   | 8.1   | 5.9   | 5.1   | 5.3   |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.213 | 0.282 | 0.313 | 0.342 | 0.346 | 0.345 | 0.353 | 0.340 | 0.323 | 0.301 | 0.345 | 0.366 |
|                                                                                  | taud        |        | 1.961 | 2.160 | 2.208 | 2.297 | 2.415 | 2.470 | 2.470 | 2.496 | 2.535 | 2.409 | 2.604 | 2.625 |
|                                                                                  | Ebn at Noon |        | 320   | 612   | 761   | 829   | 864   | 872   | 853   | 831   | 771   | 609   | 308   | 140   |
|                                                                                  | Edh at Noon |        | 25    | 59    | 87    | 101   | 99    | 96    | 93    | 83    | 65    | 47    | 22    | 14    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 0.12  | 0.57  | 1.82  | 3.65  | 4.62  | 4.76  | 4.55  | 3.48  | 1.93  | 0.80  | 0.20  | 0.05  |
|                                                                                  | RadStd      |        | 0.01  | 0.10  | 0.30  | 0.36  | 0.32  | 0.59  | 0.56  | 0.39  | 0.22  | 0.09  | 0.03  | 0.00  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (36 neighbors) | +0.47 | N/A     | N/A    | N/A     | N/A   | N/A   | -142        | -178    | N/A     | N/A     |

Nomenclature: See separate page